THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Curtains and carpets in a cinema hall are used to: 1

(a) increase echo **(b)** reflect sound
(c) absorb sound and reduce reverberation **(d)** stop light

2. The SI unit of frequency is the: 1

(a) decibel **(b)** metre **(c)** hertz **(d)** second

3. The speed of sound in air at room temperature is about: 1

(a) 34 m/s **(b)** 3 x 10^8 m/s **(c)** 1500 m/s **(d)** 344 m/s

4. The speaking tube and stethoscope work on: 1

(a) multiple reflections of sound **(b)** dispersion
(c) diffraction only **(d)** refraction of light

5. The relation between frequency (f) and time period (T) is: 1

(a) f = 2T **(b)** f = T squared **(c)** f = T **(d)** f = 1/T

6. The speed of sound in air increases with: 1

(a) frequency **(b)** temperature
(c) loudness **(d)** decrease in temperature

7. A sound wave of frequency 200 Hz and wavelength 1.7 m has a speed of: 1

(a) 117.6 m/s **(b)** 340 m/s **(c)** 201.7 m/s **(d)** 0.0085 m/s

8. The relation between speed (v), frequency (f) and wavelength (lambda) is: 1

(a) v = f + lambda **(b)** v = f/lambda **(c)** v = f lambda **(d)** v = lambda/f

9. The minimum time gap between the original sound and its echo for the echo to be heard distinctly is: 1

(a) 1 s **(b)** 0.1 s **(c)** 0.01 s **(d)** 10 s

10. The time period of a wave of frequency 50 Hz is: 1

(a) 50 s **(b)** 2 s **(c)** 0.02 s **(d)** 0.2 s

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is reverberation? How is it reduced in a hall? 2

12. Write the relation between frequency and time period. If f = 5 Hz, find T. 2

13. Why can astronauts not talk directly to each other on the Moon? 2

14. Find the wavelength of a sound wave of frequency 172 Hz in air (v = 344 m/s). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A person hears an echo after 3 s. Find the distance of the reflecting surface (v = 344 m/s). 3

16. How does a stethoscope work? 3

17. Explain why the sound of a bomb is heard louder near and fainter far away. 3

OR

Explain how bats use echolocation to catch insects in the dark. 3

18. List three applications of ultrasound. 3

OR

State three differences between sound and light. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the human ear: outer ear, middle ear and inner ear, and the path of sound to the brain. 5

OR

Explain the audible range, infrasonic and ultrasonic waves with examples of animals and uses (elephants, whales, bats, dogs). 5

20. Describe the use of ultrasound in SONAR, medical imaging and industry, and explain how a sonar finds the depth of a sea. 5

OR

Explain noise pollution, its sources, effects and control measures. 5

END OF QUESTION PAPER — Total: 40 Marks

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